Document Identifier: **PMC8047942**

SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IRB: Materials and methods Subjects and tissue specimens The Ethics Committee of The Fourth Hospital of Hebei Medical University (Shijiazhuang, China) approved this study (approval no. 2019053).
Consent: Written informed consent was obtained from all subjects or guardians.
IACUC: All animal experiments were performed with approval from the Animal Care Committee of The Fourth Hospital of Hebei Medical University (approval. no. 110324200104285428).
Euthanasia Agents: Following which, the mice were sacrificed by spinal dislocation on day 32.
Inclusion and Exclusion Criteria
not detected.
Attrition
not detected.
Sex as a biological variable
not detected.
Subject Demographics
Weight: Tumour xenograft in animals A total of 12 male athymic Balbc nude mice (4-6 weeks, 20-25 g) were purchased from Beijing Vital River Laboratory Animal Technology Co., Ltd.
Randomization
A total of three medium magnification fields (magnification, x200 light microscope) were randomly selected, and the number of migrated or invasive cells was counted.
Blinding
not detected.
Power Analysis
not detected.

Replication
qPCR was performed in triplicate with a 1:4 dilution of cDNA using the SYBR-Green PCR Kit (Promega Corporation) with the 7500 Real-Time PCR System (Applied Biosystems Thermo Fisher Scientific, Inc.).
Number: qPCR was performed in triplicate with a 1:4 dilution of cDNA using the SYBR-Green PCR Kit (Promega Corporation) with the 7500 Real-Time PCR System (Applied Biosystems Thermo Fisher Scientific, Inc.).
Cell Line Authentication
not detected.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
After blocking with normal goat serum (reagent II, streptavidin-avidin-biotin detection system cat. no. SP-9001, ZSGB-BIO) at 25degC for 30 min, sections were incubated with primary antibodies specific for FLOT-1 (1:100 cat. no. 18634 Cell Signaling Technology, Inc.), E-cadherin (1:100 cat. no. ab40772 Abcam), Ki-67 (1:100 cat. no. 27309-1-AP ProteinTech Group, Inc.), N-cadherin (1:50 cat. no. 22018-1-AP ProteinTech Group, Inc.) and Vimentin (1:100 cat. no. 10366-1-AP ProteinTech Group, Inc.) overnight at 4degC, Next, sections were incubated with the secondary antibody (reagent III, streptavidin-avidin-biotin detection system cat. no. SP-9001, ZSGB-BIO) at 37degC for 30 min, followed by incubation with a HRP-labelled streptavidin solution (reagent IV, streptavidin-avidin-biotin detection system cat. no. SP-9001, ZSGB-BIO) at 25degC for 30 min.	E-cadherin		
	Ki-67		
	N-cadherin		
The membrane was incubated with antibodies against FLOT-1 (1:1,000 cat. no. 18634 Cell Signaling Technology, Inc.), E-cadherin (1:1,000 cat. no. ab40772 Abcam), ZEB1 (1:500 cat. no. 21544-1-AP ProteinTech Group, Inc.), N-cadherin (1:500 cat. no. 22018-1-AP ProteinTech Group, Inc.), Vimentin (1:500 cat. no. 10366-1-AP ProteinTech Group, Inc.) and GAPDH (1:10,000 cat. no. AP0063 Bioworld Technology, Inc.) at 4degC for 12 h.	FLOT-1		
	ZEB1		
	Vimentin		
	GAPDH		
Experimental Models: Cell Lines			
Cell lines and culture ESCC cell lines Eca 109, KYSE 170, KYSE 180 and TE-1 were obtained from the Shanghai Institutes for Biological Sciences.	ESCC		Suggestion: (RRID:CVCL_R804)(link)
KYSE 30, KYSE 510, TE-12 and YES-2 cell lines were provided by Professor Masatoshi Tagawa (Department of Molecular Biology	YES-2		Suggestion: (RRID:CVCL_E322)(link)

and Cancer Biology, Chiba University, Chiba, Japan).			
Harvesting and subculturing of KYSE 30, Eca 109 and TE-1 cells in 96-well plates (2,500 cellswell) were then performed.	TE-1		Suggestion: RRID:CVCL_1759(link)
293T cells (Procell Life Science & Technology Co., Ltd.) were co-transfected with pmirGLO, FLOT-1 WT or FLOT-1 MUT and miR-485-5p mimics or mimics-NC with Lipofectamine(r) 2000.	293T		Suggestion: RRID:CVCL_0063(link)
Eca 109 cells were stably transfected with lentivirus-miR-485-5p or lentivirus-vector.	Eca 109		
Experimental Models: Organisms/Strains			
Tumour xenograft in animals A total of 12 male athymic Balbc nude mice (4-6 weeks, 20-25 g) were purchased from Beijing Vital River Laboratory Animal Technology Co., Ltd.	Balbc nude		
Recombinant DNA			
Transfection KYSE 30, Eca 109 and TE-1 cell lines (at 70 density), were transfected with 100 nM miR-485-5p mimics (cat. no. miR10002175-1-5 Guangzhou RiboBio Co., Ltd.), miRNA mimic negative controls (mimics-NC cat. no. miR1N0000001-1-5 Guangzhou RiboBio Co., Ltd.), miR-485-5p inhibitor (cat. no. miR20002175-1-5 Guangzhou RiboBio Co., Ltd.), miRNA inhibitor negative controls (inhibitor-NC cat. no. miR2N0000001-1-5 Guangzhou RiboBio Co., Ltd.), FLOT-1 plasmid (2 ugwell YouBio) and pcDNA 3.1 (2 ugwell YouBio) using Lipofectamine(r) 2000 (Invitrogen Thermo Fisher Scientific, Inc.), according to the manufacturers instructions.	FLOT-1		
The wild-type (WT) or mutant (MUT) miR-485-5p binding sequence of FLOT-1 was inserted into the pmirGLO plasmid (YouBio) to establish the recombinant luciferase reporter plasmids (FLOT-1 WT and FLOT-1 MUT).	pmirGLO		
Software and Algorithms			
Statistical analysis Statistics were calculated using GraphPad Prism	GraphPad Prism		

(v5.0) software (GraphPad Software, Inc.) and SPSS v22.0 (IBM Corp.).	GraphPad		Suggestion: GraphPad Prism, RRID:SCR_002798(link)
	SPSS		Suggestion: SPSS, RRID:SCR_002865(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
For comparison among the paired samples, paired t-tests were performed.	t-tests
For comparison among the unpaired samples, an unpaired Students t-test was performed.	t-test
One-way ANOVA followed by Dunnetts post hoc test was used to examine the significant differences in the experimental data between multiple groups.	ANOVA
	Dunnetts post hoc test
The relationship between clinicopathological results and the expression of miR-485-5p was examined using a Pearsons chi-squared test.	chi-squared test
The correlation between miR-485-5p and FLOT-1 expression was analysed using Spearmans correlation analysis.	Spearmans correlation
Oligonucleotides	
Gene specific qPCR primers were as follows: U6 snRNA reverse transcription primer, 5-CGCTTCACGAATTTGCGTGTCAT-3 U6 snRNA forward, 5-GCTTCGGCAGCACATATACTAAAAT-3 and reverse, 5-CGCTTCACGAATTTGCGTGTCA T-3 miR-485-5p reverse transcription primer, 5-GTCGTATCCAGTGCGTGTCGTGGAGTCGGCAATTGCACTGGATACGACGAATTCA-3 miR-485-5p forward, 5-GGAGAGGCTGGCCGTGAT-3 and reverse, 5-CAGTGCGTGTCGTGGAGT-3 N-cadherin forward, 5-TGTTTGACTATGAAGGCAGTGG-3 and reverse, 5-TCAGTCATCACCTCCACCAT-3 E-cadherin forward, 5-CCGCCGGCGTCTGTAGGAA-3 and reverse, 5-AGGGCTCTTTGACCACCGCTCTC-3 Vimentin forward, 5-GAGAACTTTGCCGTTGAAGC-3 and reverse, 5-TCCAGCAGCTTCCTGTAGGT-3 zinc finger E-box-binding homeobox 1 (ZEB1) forward, 5-TTGTAGCGACTGGATTTT-3 and reverse, 5-AGACGATAGTTGGGTCCCGGC-3 FLOT-1 forward, 5-CCCATCTCAGTCACTGGCATT-3 and reverse, 5-CCGCCAACATCTCCTTGTTC-3 and GAPDH forward, 5-GGACCTGACCTGCCGTCTAG-3	5-GAGAACTTTGCCGTTGAAGC-3

and reverse, 5-GTAGCCCAGGATGCCCTTGA-3 .	

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq